

Dilip Kumar

AI/ML ENGINEER · ASPIRING AI RESEARCHER & MODEL TRAINER

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SUMMARY

Second-year B.Tech student who ships AI systems end to end. I wrote an offline voice-typing app for Windows in Rust that runs Whisper on your own machine, and a multi-agent platform in Python with a React dashboard on top of it. What I actually care about is the training side: building the datasets, running the fine-tunes, and pre-training a small language model from scratch so I understand what everyone else is importing.

SKILLS

Languages	Python · Rust · TypeScript · C/C++ · SQL
ML / Deep learning	PyTorch · TensorFlow · scikit-learn · NumPy · Pandas · OpenCV
GenAI / LLMs	Transformers · Hugging Face · LangChain / LangGraph · RAG + vector DBs · QLoRA · Whisper (on-device)
Data & training	Dataset curation · Tokenizers · Eval harnesses · Synthetic data
Frameworks & cloud	React 19 · Tauri 2 · Docker · FastAPI · AWS · Linux · CI/CD
Security	Kali Linux · Burp Suite · OWASP Top 10 · TryHackMe

PROJECTS

Bolo · Offline voice typing for Windows Rust, Tauri 2, whisper.cpp · Shipped · v0.10 · MIT

Built an offline push-to-talk dictation app for Windows in Rust and Tauri 2, running whisper.cpp in-process so no audio ever leaves the machine. Cut word error rate from 13.2% to 9.6% on a 41-clip test set by adding a personal-vocabulary dictionary; 5.0% on clean English, ~1.9 s p50 to text on laptop CPU.
Gated every release behind an evaluation harness, and shipped 10 public versions with a 3 MB installer.
github.com/Dilip-kumar-22/bolo

Typing Master · Touch-typing tutor, installable and offline TypeScript, Preact, PWA · Live · v2.1 · MIT

Shipped an installable, offline-first typing tutor: 41-chapter curriculum, adaptive drills weighted to the keys the user fails, colour-coded finger guide and a 3D keyboard heatmap.
Kept all progress on-device with no account, and added multiplayer races and timed challenges.
dilip-kumar-22.github.io/typing-master-scorp/

S-CORP · A multi-agent platform shaped like a company Python, React 19, TypeScript · Private beta

Architected a multi-agent AI platform where agents hold defined roles and hand work to each other, on a Python backend.
Built a React 19 command dashboard for live control, and split the system into a three-repo monorepo (core / backend / frontend).

ORBIT · 3D scrollytelling starter, given away Three.js, GLSL, OKLCH · Live · Open source · MIT

Released an open-source 3D scrollytelling starter with HDR-safe post-processing (HalfFloat + 4× MSAA), an OKLCH palette shared between CSS and WebGL, and reduced-motion and no-WebGL fallbacks built in.
dilip-kumar-22.github.io/orbit/

FRIDAY · Terminal-native assistant, 13 sub-agents Python, Gemini · Private

Built a terminal-native assistant from 13 specialised sub-agents over a three-tier memory architecture.

RESEARCH & MODEL TRAINING

Pre-training a custom ~200M-parameter language model from scratch: tokenizer, training loop and evaluation.
Fine-tuning 70B-class open models with QLoRA on curated domain data.
Dataset curation and eval-first pipeline design: deduplication, filtering, and writing the evaluation before the model.
One arXiv paper a week, re-implemented in PyTorch.

EDUCATION

B.Tech, Computer Science & Engineering (AI & ML)

Lovely Professional University, Punjab, India
2025 – 2029 (expected) · CGPA 9.00

CERTIFICATIONS

AWS Cloud Practitioner — in progress, 2026
Targeted next: AWS Solutions Architect Associate (2027),
DeepLearning.AI Deep Learning Specialization (2027), AWS
ML Specialty (2028).